REFONTE INTERNATIONAL TRAINING AND INTERNSHIP PROGRAM

Astrodynamics Specialist

SESSION 4

Interplanetary Transfers and Gravity Assists

From Earth to Mars, and Slingshots Around Planets

Prepared for Cao Khanh Tan

Module 4, closing the Core Training Phase

Builds on Sessions 1 to 3: vis-viva, Hohmann transfers, hyperbolic orbits

Tools poliastro (ephem, Lambert), GMAT, STK Astrogator

SESSION 4

Interplanetary Transfers and Gravity Assists

A Hohmann transfer between two circular orbits is exactly the tool we need to go from Earth to Mars, except now the central body is the Sun and the two orbits belong to moving planets. This module scales the delta-v thinking of Session 3 up to the Solar System. We break the journey into simple two-body pieces with the patched-conic method, size the Earth-to-Mars transfer named on the programme page, work out the departure and arrival burns, find when you are actually allowed to launch, and finish with the one manoeuvre that gives energy for free: the gravity assist.

Learning objectives

- Explain the patched-conic method and the role of a planet sphere of influence.
- Size a heliocentric Hohmann transfer and read off the hyperbolic excess speeds at each end.
- Compute the departure (TMI) and arrival (MOI) burns from and to parking orbits.
- Define C3 and use the synodic period and phase angle to find a launch window.
- Explain why a gravity assist changes heliocentric speed without spending propellant.
- Compute a flyby turning angle and the velocity change it produces.

> **The whole trip in one sentence**
> Escape Earth onto a Sun-centred transfer ellipse, coast to the target planet as it arrives at the same point, then capture into orbit. Everything below is the delta-v bookkeeping for those three steps.

PART 1

The Patched-Conic Method

Solving the full three-or-more-body problem exactly is hard. The patched-conic approximation replaces it with a chain of familiar two-body problems, patched together at the boundaries where the dominant gravity changes hands:

- **Near Earth:** a hyperbolic escape orbit, governed by Earth gravity.
- **Between planets:** a heliocentric transfer ellipse, governed by the Sun.
- **Near the target:** a hyperbolic arrival orbit, governed by the target planet.

The handoff happens at the sphere of influence (SOI), the radius inside which a planet gravity dominates the Sun. Its size is:

$$r_{\mathrm{SOI}} = a_p \left(\frac{m_p}{m_\odot} \right)^{2/5}$$

Earth SOI is about 924,000 km and Mars about 577,000 km. These are tiny compared to the 1.5 AU (225 million km) between the planets, so from the Sun point of view the escape and arrival hyperbolas look like single points: the spacecraft simply leaves Earth with some excess velocity and arrives at Mars with another. That excess speed, the hyperbolic excess velocity v-infinity, is the quantity that stitches the conics together.

PART 2

The Heliocentric Transfer: Earth to Mars

Treating the planet orbits as circular and coplanar, the minimum-energy route is a Sun-centred Hohmann ellipse tangent to Earth orbit at perihelion and Mars orbit at aphelion:

$$a_t = \frac{r_1 + r_2}{2}, \qquad t_{\text{transfer}} = \pi\sqrt{\frac{a_t^3}{\mu_\odot}}$$

Earth moves at 29.8 km/s and Mars at 24.1 km/s around the Sun. On the transfer ellipse the spacecraft needs 32.7 km/s at the Earth end and 21.5 km/s at the Mars end. The differences from the local planet speed are the excess velocities that the escape and arrival hyperbolas must supply:

$$v_{\infty,\,\text{dep}} = v_{p,\,\text{helio}} - v_1$$

$$v_{\infty,\,\text{arr}} = v_2 - v_{a,\,\text{helio}}$$

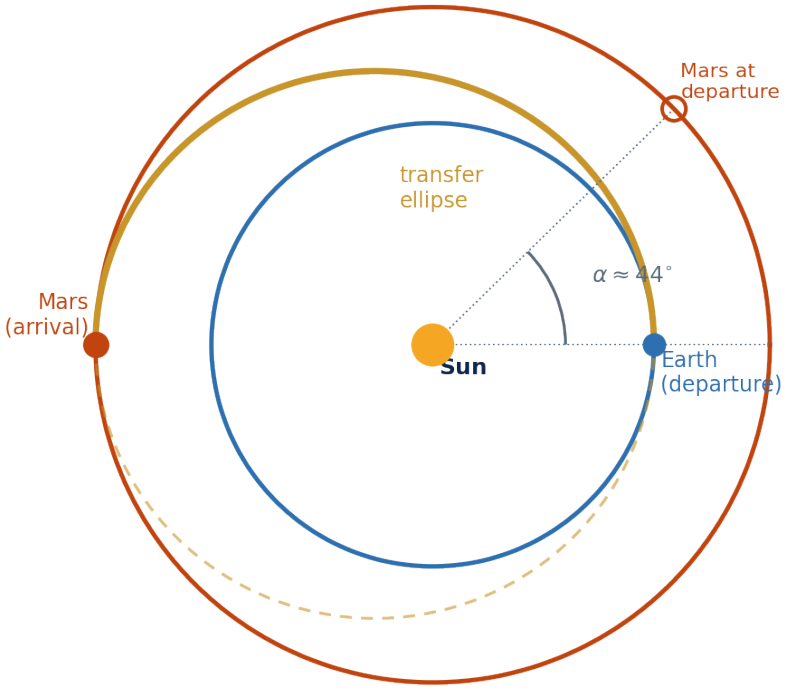


Figure 1. The Earth-to-Mars patched-conic transfer. The spacecraft departs Earth, coasts along the Sun-centred transfer ellipse (gold), and meets Mars at aphelion. For that rendezvous to work, Mars must lead Earth by a specific phase angle at departure, shown by the open circle.

Worked example: Earth to Mars, heliocentric leg

The transfer ellipse has a = 1.26 AU. At the Earth end the excess speed is v-infinity = 32.73 - 29.79 = 2.94 km/s; at Mars it is 24.13 - 21.48 = 2.65 km/s. The coast is half the ellipse period, about 259 days, or 8.5 months. Those two excess speeds are the inputs to the departure and arrival burns in Part 3.

PART 3

Departure and Arrival Hyperbolas

The spacecraft does not start in heliocentric space; it starts in a parking orbit around Earth. To leave with excess speed v-infinity it must reach, at the parking radius, the speed at the perigee of an escape hyperbola:

$$v_p = \sqrt{v_\infty^2 + \frac{2\mu_p}{r_p}}$$

The trans-Mars injection burn (TMI) is the jump from circular parking speed to that hyperbolic perigee speed:

$$\Delta v_{\text{TMI}} = v_p - v_c = \sqrt{v_\infty^2 + \frac{2\mu}{r_p}} - \sqrt{\frac{\mu}{r_p}}$$

The launch energy is often quoted as C3, the square of the excess velocity, because it is what a launch vehicle brochure lists:

$$C_3 = v_\infty^2$$

Worked example: the two burns
From a 300 km LEO (v_circ = 7.73 km/s), reaching v-infinity = 2.94 km/s needs a hyperbolic perigee speed of 11.32 km/s, so TMI = 3.59 km/s, at a launch energy C3 = 8.67 km^2/s^2. Arriving at Mars with v-infinity = 2.65 km/s, capturing into a low 500 km circular orbit costs MOI = 2.07 km/s. Capturing instead into a wide ellipse (500 x 20,000 km) costs only about 1.04 km/s, and aerobraking can cut it further, which is why real Mars orbiters rarely circularise propulsively.

PART 4

Launch Windows

You cannot launch to Mars whenever you like. The spacecraft and Mars must arrive at the aphelion point together, so Mars has to be ahead of Earth by just the right angle at departure. Since Mars moves during the 259-day coast, the required lead (phase) angle is:

$$\alpha = 180^\circ - n_2\, t_{\text{transfer}}$$

For Earth to Mars this works out to about 44 degrees. That geometry only repeats once per synodic period, the time for the two planets to return to the same relative configuration:

$$\frac{1}{S} = \left| \frac{1}{T_1} - \frac{1}{T_2} \right|$$

which for Earth and Mars is about 780 days, roughly 26 months. That is why Mars missions launch in clusters every two years or so. Real mission design replaces the circular-coplanar assumption with a Lambert solver over a grid of departure and arrival dates, producing the familiar porkchop plot of C3 contours; the Hohmann result sits at the bottom of that plot as the cheapest, slowest option.

PART 5

Gravity Assists

A gravity assist steals a little of a planet orbital momentum to change the spacecraft heliocentric velocity for free. The trick lives in the change of reference frame. In the planet own frame, the flyby is just a hyperbola: the spacecraft comes in and leaves with the same excess speed, but its direction is rotated by the turning angle:

$$\sin\frac{\delta}{2} = \frac{1}{e}, \qquad e = 1 + \frac{r_p\, v_\infty^2}{\mu_p}$$

In the planet frame the flyby leaves the speed | v_inf | unchanged and only turns its direction by the angle delta. In the Sun frame that rotation changes the heliocentric velocity from V_in to V_out: here a longer vector, a free boost.

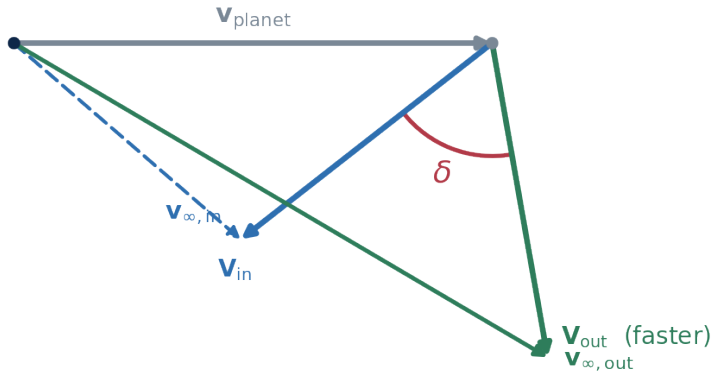


Figure 2. Why a flyby gives a free boost. In the planet frame only the direction of v-infinity changes. Adding the planet velocity back in, the heliocentric velocity swings from V_in to V_out, which here is longer: the spacecraft has sped up relative to the Sun without firing its engine.

The magnitude of the heliocentric velocity change from a single flyby is:

$$|\Delta \mathbf{v}_{\text{helio}}| = 2\, v_\infty \sin\frac{\delta}{2}$$

A closer, slower flyby bends the path more and delivers a bigger boost, as Figure 3 shows. Passing behind the planet speeds the craft up; passing in front slows it down, which is how missions to the inner Solar System shed energy.

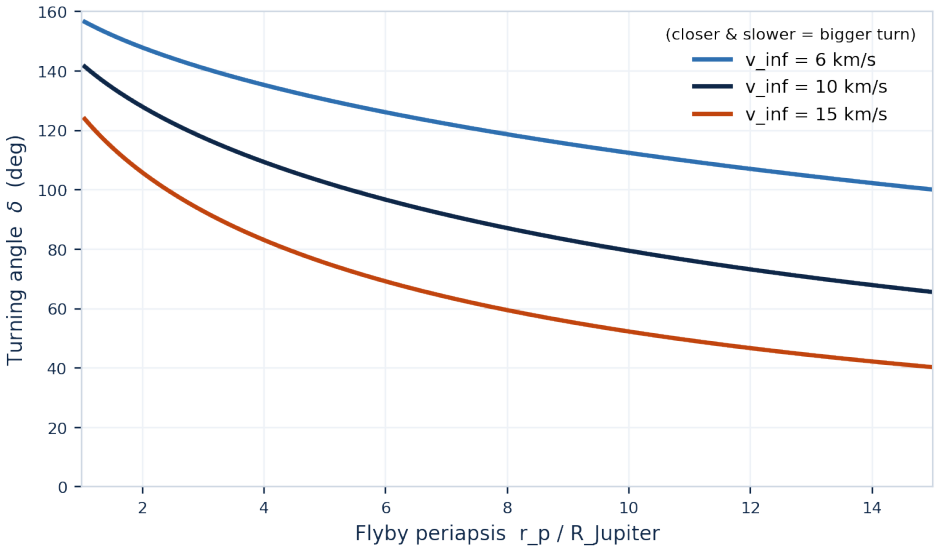


Figure 3. Turning angle versus flyby periapsis for a Jupiter flyby at three excess speeds. Lower and slower means a larger turn. Jupiter, with the deepest gravity well of the planets, is the great accelerator of the outer Solar System.

Worked example: a Jupiter flyby

A spacecraft passing Jupiter at v-infinity = 10 km/s with a periapsis of 3 Jupiter radii has eccentricity e = 1.17 and a turning angle of about 118 degrees. The resulting heliocentric velocity change can be as large as 2 v-infinity sin(delta/2) = 17 km/s, far more than any chemical burn a probe could carry. This is how the Voyagers reached the outer planets.

Computing Interplanetary Transfers

The heliocentric Hohmann and the hyperbolic burns are a few lines of vis-viva, now with the Sun and Earth gravitational parameters. The comment shows the real-world tools that replace the circular-coplanar assumption.

```
import numpy as np
mu_sun, mu_e = 1.32712440e11, 398600.4418
AU = 1.495978707e8
rE, rM = 1.000*AU, 1.523679*AU

vis = lambda r,a,mu: np.sqrt(mu*(2/r - 1/a))
vc  = lambda r,mu: np.sqrt(mu/r)
a_t = (rE + rM)/2

vinf_dep = vis(rE,a_t,mu_sun) - vc(rE,mu_sun)   # ~2.94 km/s
vinf_arr = vc(rM,mu_sun) - vis(rM,a_t,mu_sun)   # ~2.65 km/s
t_days   = np.pi*np.sqrt(a_t**3/mu_sun)/86400   # ~259 days

rp = 6378.137 + 300                              # 300 km LEO
tmi = np.sqrt(vinf_dep**2 + 2*mu_e/rp) - vc(rp,mu_e)   # ~3.59 km/s

# Real design: poliastro Lambert / ephemerides, or a GMAT / STK
# Astrogator porkchop scan over departure and arrival dates.
```

Production tools

poliastro provides Lambert solvers and planetary ephemerides to build real transfers between dated positions; GMAT and STK Astrogator add porkchop scans, patched-conic and full-force targeting, and gravity-assist tour design. The hand calculations here are the sanity check you run before trusting any of them.

PRACTICE

This Week (aim for 12 to 15 hours)

Bring your work and plots to the next live session. As always, show your reasoning, not just the final numbers.

Assignment: Interplanetary Transfers and Gravity Assists

- **Reading and notes (3 to 4 h).** Explain the patched-conic method in your own words, and why the excess speed v-infinity is the same going into and coming out of a gravity assist. One page.
- **Inner transfer (3 to 4 h).** Size a heliocentric Hohmann from Earth to Venus (0.723 AU). Report both excess speeds and the transfer time. Note that here you slow down to fall inward.
- **Full Earth-to-Mars budget (4 to 5 h).** From a 400 km LEO, compute TMI and C3, then the MOI to capture into a 500 x 20,000 km Mars ellipse. Report the total propulsive delta-v.
- **Gravity assist (2 h).** For a Jupiter flyby at v-infinity = 8 km/s with periapsis 4 Jupiter radii, compute the eccentricity, the turning angle, and the maximum heliocentric velocity change.
- **STK or GMAT (2 h).** Build an Earth-to-Mars transfer (Astrogator or a GMAT interplanetary sequence) and compare its transfer time and departure C3 to your hand values.

Answer key (for your self-check)

Quantity	Expected value
Earth to Mars, heliocentric	v_inf 2.94 (dep) and 2.65 (arr) km/s; ~259 days
Earth to Venus, heliocentric	v_inf 2.50 (dep) and 2.71 (arr) km/s; ~146 days
TMI from 400 km LEO	about 3.57 km/s; C3 = 8.67 km^2/s^2
MOI into 500 x 20,000 km ellipse	about 1.04 km/s; total mission ~4.61 km/s
Jupiter flyby, v_inf 8, r_p 4 R_J	e = 1.14, turning ~122 deg, max dv ~14.0 km/s
Earth-Mars synodic / phase	~780 days; Mars leads by ~44 deg

Coming up in Session 5

Satellite constellations and coverage: Walker constellations, revisit time and ground-track geometry, and how many satellites it takes to see the whole Earth. Back in Earth orbit, and straight from the programme constellation-design project.

REFERENCE

One-Page Formula Sheet

Sphere of influence	$r_{\mathrm{SOI}} = a_p \left(\frac{m_p}{m_\odot}\right)^{2/5}$
Transfer ellipse and time	$a_t = \frac{r_1 + r_2}{2}, \qquad t_{\mathrm{transfer}} = \pi \sqrt{\frac{a_t^3}{\mu_\odot}}$
Excess speed at departure	$v_{\infty,\,\mathrm{dep}} = v_{p,\,\mathrm{helio}} - v_1$
Excess speed at arrival	$v_{\infty,\,\mathrm{arr}} = v_2 - v_{a,\,\mathrm{helio}}$
Hyperbola periapsis speed	$v_p = \sqrt{v_\infty^2 + \frac{2\mu_p}{r_p}}$
Injection burn (TMI)	$\Delta v_{\mathrm{TMI}} = v_p - v_c = \sqrt{v_\infty^2 + \frac{2\mu}{r_p}} - \sqrt{\frac{\mu}{r_p}}$
Launch energy	$C_3 = v_\infty^2$
Synodic period	$\frac{1}{S} = \left\lvert \frac{1}{T_1} - \frac{1}{T_2} \right\rvert$
Phase angle	$\alpha = 180^\circ - n_2\, t_{\mathrm{transfer}}$
Flyby turning angle	$\sin\frac{\delta}{2} = \frac{1}{e}, \qquad e = 1 + \frac{r_p\, v_\infty^2}{\mu_p}$
Assist velocity change	$\lvert \Delta \mathbf{v}_{\mathrm{helio}} \rvert = 2\, v_\infty \sin\frac{\delta}{2}$

Key glossary

- **Patched conic:** stitching Earth, Sun and target two-body arcs together at the spheres of influence.
- **v-infinity:** hyperbolic excess speed; the leftover speed after escaping a planet gravity. Its square is C3, the launch-energy figure of merit.
- **Synodic period:** the time between identical planet-to-planet geometries; sets the launch window.
- **Gravity assist:** a flyby that rotates v-infinity in the planet frame, changing heliocentric speed for free.

Going further

For depth, see Curtis (interplanetary trajectories and flybys), Vallado (Lambert and mission design), and Bate, Mueller and White (the patched-conic classics). These are real, widely used texts; please confirm editions yourself, as I cannot verify those details from here.